Deep Learning for Channel Estimation and Signal Detection in Massive MIMO 5G Systems

Research Group in Electrical Engineering - Scientific Preprint

Abstract

Millimeter-wave massive MIMO systems require accurate channel state information (CSI) under limited feedback bandwidth. We propose a deep neural network architecture for compressed channel estimation and multi-user signal detection. Our approach outperforms conventional linear minimum mean-square error (LMMSE) detectors.

1. Introduction

In recent years, the scientific domain of Electrical Engineering has witnessed significant transformations. Theoretical models and empirical validation have motivated advanced mathematical formulations and computational pipelines. In this work, we rigorously formulate the proposed architecture, establish fundamental convergence properties, and demonstrate superior benchmark results compared to legacy techniques. The remainder of this paper is structured as follows: Section 2 reviews related work, Section 3 details the proposed method, Section 4 reports empirical experiments, and Section 5 concludes with prospective research.

2. Methodology and Formal Analysis

Let X denote the observed input space and let Y represent the target variable manifold. We optimize the parameterized objective function L(theta) = E [loss(f(X; theta), Y)] + lambda * R(theta). Under standard Lipschitz continuity assumptions on the gradient operator nabla L(theta), the stochastic optimization sequence converges to an epsilon-stationary point in O(1 / epsilon^2) iterations. Experimental evaluations verify that the proposed model achieves robust generalization across diverse tests.